

Funding Rates and the Conditional Informativeness of Order Flow:

Evidence from the Binance XAUUSD Gold Perpetual

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Abstract

One of the key institutions in a perpetual futures contract is the funding-rate mechanism, yet its role in information processing remains largely unexplored. This paper examines the question using the Binance XAUUSD gold perpetual, a commodity-linked cryptocurrency perpetual introduced in January 2026, and 203,037 one-minute observations from its early trading period. In this benchmark-anchored instrument, whose reference value is discovered off-exchange, the order-flow proxies used in the paper have no unconditional predictive ability: trade-based order-flow imbalance is insignificant in-sample, produces negative out-of-sample predictive R-squared, and fails to survive multiple-testing corrections. The active channel is instead the funding mechanism. The funding rate positively predicts returns at five-, fifteen-, and thirty-minute horizons ($t = 2.22, 2.95, 3.71$), while the funding-by-order-flow interaction is negative and significant at all three horizons ($t = -2.64, -2.28, -2.26$). The positive funding coefficient captures the persistence and directional component of leverage-driven demand, whereas the negative interaction captures the marginal reversal effect of additional buying pressure when positioning is already crowded. Session-level patterns around the European window are stable in sign but do not survive multiple-testing correction, alternative window definitions, alternative normalisations, or non-overlapping sampling schemes; event-study and placebo evidence do not isolate the London afternoon fix as the source of the pattern. The effects are small and not profitable once trading costs are considered. The evidence points to a funding-based positioning-and-reversal channel rather than a classical informed-trading channel in commodity-linked perpetuals.

Keywords: *funding rate; perpetual futures; commodity-linked derivatives; order-flow imbalance; positioning; information transmission; cryptocurrency market microstructure; gold.*

JEL Classification: *G12, G13, G14, G15, C58.*

Declaration of Generative AI and AI-Assisted Technologies. *During the preparation of this work, the author used AI-assisted tools for language editing, code-review support, and organisation of exposition. After using these tools, the author reviewed and edited the content and takes full responsibility for the content of the paper. All data collection, empirical design, estimation, and interpretation were conducted and verified by the author.*

1. Introduction

The defining feature that separates a perpetual futures contract from every dated future is that it never expires, and is instead tethered to its reference price by a funding payment exchanged between longs and shorts. This mechanism has become the central price-anchoring device of crypto derivatives, but its role in the transmission of information into prices has not been studied. The question becomes especially sharp in a new instrument: the commodity-linked perpetual — a stablecoin-settled, continuously traded contract referencing an externally priced commodity benchmark rather than a crypto-native asset, of which the Binance XAUUSDT gold perpetual (launched January 2026) is the leading example. In such a contract the fundamental price is discovered off-exchange, in the deep over-the-counter and futures markets for gold. We therefore ask not merely whether exchange-level order flow predicts returns, but which channel carries return-relevant information when fundamental value is set elsewhere: raw trade flow, as classical microstructure would predict, or the funding mechanism that prices leveraged positioning.

Three design features make this a question about market architecture rather than a routine predictability test. First, the contract trades 24 hours a day, but its gold index is live only during gold-market hours and is held fixed otherwise, with a smoothed off-hours mark price — so the very availability of fundamental information varies within the trading day. Second, funding is settled every four hours and summarises the net positioning pressure in the contract; it is an internal, near-continuous restoring force with no analogue in equities or dated futures, and it prices the state of leveraged demand directly. Third, the information events most relevant for gold — the twice-daily London (LBMA) fixes — fall on an external institutional calendar that maps onto only part of the trading day. Together these imply that, in a benchmark-anchored perpetual, exchange-level order flow may convey little about gold's value, while the funding mechanism may carry the information that flow does not. The contract's design, in other words, predicts where information should and should not appear — a hypothesis we take directly to the data using 203,037 one-minute observations covering the contract from launch through 21 May 2026.

The evidence supports this architecture in two mutually reinforcing parts. The first is a disciplined null: raw minute-level trade-based order-flow imbalance has no unconditional predictive power for short-horizon returns — insignificant in-sample, negative out-of-sample R-squared, and not surviving a multiple-testing correction across the fifteen session-by-horizon cells we examine. Rather than a disappointment, this is consistent with what a benchmark-anchored design would imply: when fundamental value is discovered off-exchange, on-exchange flow should carry limited standalone information about it. The second part is the paper's central result and identifies the active channel. Order-flow informativeness is conditional on the funding regime: the imbalance-by-funding interaction is negative and significant at all three horizons ($t = -2.64, -2.28, -2.26$), and the funding rate is itself a positive and significant predictor at every horizon ($t = 2.22, 2.95, 3.71$) and the single most robust relationship in the study. These two coefficients are not in tension but capture distinct forces. The positive funding main effect reflects the persistence of the funding state and the directional component of leveraged demand — a high funding rate tends to accompany subsequent positive returns. The negative interaction captures a marginal effect of a different kind: once that state is already crowded (high funding), an additional minute of

buying pressure is more likely to represent crowded, over-extended positioning, and is therefore followed by a reversal rather than continuation. The resulting mechanism is positioning-and-reversal — exchange flow conveys information about the state of leveraged positioning, which funding prices — rather than classical informed trading about gold value. Consistent with the same architecture, session-level effects around the European window are directionally stable but fragile (they do not survive multiple-testing correction, alternative session boundaries, alternative normalisations, or non-overlapping sampling), and an event study with placebo windows does not isolate the London afternoon fix; we report them as suggestive only. Every documented effect is small and not exploitable net of trading costs — we present them as evidence about market structure, not as a tradable strategy.

We make three contributions. First and most substantively, we characterise the information architecture of a benchmark-anchored perpetual: return-relevant information is transmitted through the funding mechanism and positioning dynamics rather than through raw order flow, and order-flow informativeness is conditional on the funding state. Because this links a contract’s institutional design — its funding cadence, its off-hours index/mark-price construction, and its leverage — to the way information enters prices, it speaks directly to the market design of the fast-growing commodity-linked perpetual class. Second, we provide the first high-frequency microstructure evidence on a gold-linked cryptocurrency perpetual, an instrument bridging commodity derivatives and digital-asset markets that has not previously been examined at this level. Third, we establish a robust structured null on unconditional trade-imbalance predictability — a disciplined contrast with the strong predictability documented for depth-based order-flow imbalance in equity limit order books, and itself a prediction of the benchmark-anchored architecture.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting. Section 3 reviews the related literature. Section 4 details the data and variable construction. Section 5 sets out the methodology. Section 6 presents the results. Section 7 reports robustness. Section 8 discusses implications, and Section 9 concludes.

2. Institutional Background

2.1 The Binance XAUUSDT perpetual

The XAUUSDT contract is a USDT-settled perpetual futures contract referencing the price of one troy ounce of gold, launched by Binance on 5 January 2026 and operated by Nest Exchange Limited under the regulation of the Financial Services Regulatory Authority of the Abu Dhabi Global Market (ADGM). Unlike a dated commodity future, it has no expiry; unlike physical gold, it is cash-settled in a stablecoin; and unlike the underlying spot market, it trades continuously, 24 hours a day, with leverage. Our sample therefore covers essentially the entire trading history of the instrument from inception, which we treat as early-market evidence.

2.2 The funding-rate mechanism

Because the contract never expires, its price is tethered to the gold index through a periodic funding payment exchanged between long and short holders. When the perpetual trades at a

premium to the index, longs pay shorts, and vice versa, creating a restoring force toward the benchmark. For XAUUSDT the funding rate is settled every four hours and is capped at plus or minus two percent per interval. This four-hourly cadence, rather than the eight-hourly cadence common to major crypto perpetuals, is the relevant frequency for any funding-conditioned analysis. Economically, the funding rate summarises the net positioning pressure in the contract: a persistently positive rate reflects crowded long demand, a negative rate crowded short demand.

2.3 The external benchmark and the index/mark-price construction

The contract’s index price aggregates gold quotations from multiple vendors and updates every second during gold-market hours; outside those hours the index is held fixed at its last value, while the mark price that governs liquidations is propagated by an exponentially weighted moving average to damp off-hours swings, subject to a deviation cap (illustratively plus or minus three percent for XAUUSDT). This design is central to interpreting intraday results: during gold off-hours the reference price is mechanically static, so contract-price movements and order flow in those windows reflect inventory and liquidity dynamics on the exchange rather than fresh information impounded from a live benchmark. We return to this point when interpreting the overnight session.

2.4 The LBMA gold fixes and intraday information

The global reference prices for physical gold are the twice-daily LBMA fixes, conducted at 10:30 and 15:00 London time, which concentrate institutional participation and are documented sources of short-horizon price adjustment in the traditional gold market (Caminschi and Heaney, 2014). Because the contract’s index is live during London hours, the European trading window is the period in which externally generated benchmark information is most likely to enter the perpetual. We note, however, that the London afternoon fix falls at 15:00 UTC under Greenwich Mean Time and 14:00 UTC under British Summer Time; under a fixed 08:00-14:00 UTC definition of the European session, the afternoon fix lies at or beyond the session boundary. We use this institutional timing to motivate, rather than to assert, a benchmark-timing interpretation of the session results, and we test it explicitly in Section 7.

3. Related Literature

3.1 Order flow and market microstructure

The information content of order flow is a central question in market microstructure. The sequential-trade and strategic-informed-trader models of Glosten and Milgrom (1985) and Kyle (1985) predict that net directional order flow conveys information about fundamental value, as informed agents trade in the direction of their private signals and prices adjust toward efficient levels; O’Hara (2015) synthesises how these mechanisms operate at high frequency. Consistent with the theory, signed order flow has measurable price impact and some short-horizon predictive content: Cont, Kukanov, and Stoikov (2014) show that a depth-based order-flow imbalance built from revisions in best bid and ask depth predicts short-horizon equity price changes, and Chordia, Roll, and Subrahmanyam (2002) find that aggregate order imbalance forecasts market returns and liquidity. The empirical micro-

foundations rest on trade-direction classification (Lee and Ready, 1991), on liquidity and price-impact measures (Amihud, 2002), and on order-flow-toxicity diagnostics for high-frequency markets (Easley, Lopez de Prado, and O’Hara, 2012). Two features of this evidence matter here. First, it is drawn overwhelmingly from equities, conventional dated futures, and limit-order-book venues. Second, the canonical imbalance measures are depth-based, whereas our measure is trade-based — computed from executed buyer- and seller-initiated volume rather than quote-level depth — so it captures realised directional pressure rather than book revisions, and we do not treat our coefficients as directly comparable to the depth-based OFI of Cont, Kukanov, and Stoikov (2014).

3.2 Perpetual futures and the funding-rate mechanism

Perpetual futures differ from dated futures because they have no maturity and therefore no terminal convergence through contract expiry. Their benchmark linkage is maintained instead through the funding-rate mechanism: periodic payments between long and short holders that penalise one-sided deviations from the reference price. Economically, the funding rate summarises net long-short pressure, basis pressure relative to the benchmark, and the cost of maintaining leveraged exposure, and it has become the central price-anchoring device in crypto derivatives. Existing high-frequency studies of perpetual and crypto derivatives focus primarily on Bitcoin markets: Alexander, Choi, Park, and Sohn (2020) document BitMEX’s role in bitcoin price discovery, Alexander, Heck, and Kaeck (2022) study Binance’s role in volatility transmission, and Makarov and Schoar (2020) analyse arbitrage frictions and persistent cross-platform price differences. This literature is the natural background for our setting, but it has not examined a commodity-linked perpetual whose reference value is discovered in traditional gold markets.

3.3 Crypto derivatives, basis, and price discovery

A large literature asks where price discovery occurs in fragmented, continuously trading crypto markets. Findings on spot-versus-futures leadership are mixed and time-varying: Baur and Dimpfl (2019) and Entrop, Frijns, and Seruset (2020) document shifting leadership driven by relative trading costs and volume, Alexander and Heck (2020) find that unregulated derivatives venues dominate price discovery, and Fassas, Papadamou, and Koulis (2020) study bitcoin-futures price discovery, with the common toolkit being the cointegration and information-share framework (Engle and Granger, 1987; Hasbrouck, 1995). Closest to our microstructural focus, Aleti and Mizrach (2021) examine the spot and futures market microstructure of bitcoin. Venue thinness and design materially affect these estimates (Adammer, Bohl, and Gross, 2016), and the integrity of crypto volume and pricing has itself been scrutinised (Griffin and Shams, 2020; Corbet, Lucey, Urquhart, and Yarovaya, 2019). This work is almost entirely about native crypto assets; commodity-linked crypto perpetuals are essentially unstudied.

3.4 Gold market microstructure and price discovery

For gold specifically, price discovery has traditionally been located in established venues: the OTC London market with its twice-daily LBMA benchmark fixes, and COMEX futures. Caminschi and Heaney (2014) document short-horizon information leakage around the London PM fix; Hauptfleisch, Putnins, and Lucey (2016) decompose price discovery

between London and New York; Garbade and Silber (1983) provide the classic futures-versus-cash price-discovery framework; and O’Connor, Lucey, Batten, and Baur (2015) survey the financial economics of gold. This matters because a gold-linked perpetual is anchored to a price that is largely discovered off-exchange. If the external benchmark and the funding mechanism jointly absorb fundamental information, exchange-level order flow on the perpetual may be comparatively uninformative about gold value — the opposite of what order-flow theory predicts for venues where price discovery happens on-book.

3.5 Where this paper fits

These literatures have developed largely in isolation: order flow and market microstructure in equities and dated futures; funding, basis, and price discovery in native crypto derivatives; and gold price discovery in traditional London and COMEX venues. This paper connects the three. To our knowledge, no existing study examines whether minute-level trade-based order-flow imbalance predicts short-horizon returns in a commodity-linked perpetual whose reference value is discovered off-exchange. The Binance XAUUSDT contract is a useful laboratory for this question because it combines continuous, crypto-style trading, leveraged stablecoin settlement, an external gold benchmark, and a funding mechanism that prices positioning pressure directly.

4. Data and Variable Construction

4.1 Data source and sample

Our main data are tick-level trades collected from the Binance Futures API. The API query window begins on 1 January 2026, but the usable sample starts with the launch of the XAUUSDT contract on 5 January 2026 and ends on 21 May 2026. Each tick record contains a millisecond-precision timestamp, transaction price, quantity, and a buyer/seller aggressor indicator. We aggregate the raw ticks into one-minute OHLCV bars and remove bars with missing or zero-volume observations, yielding a final sample of 203,037 one-minute observations.

4.2 Variable construction

Trade-based order-flow imbalance (OFI). We construct a trade-based order-flow imbalance measure, $OFI(t) = \text{BuyVolume}(t) - \text{SellVolume}(t)$, the difference between buyer- and seller-initiated executed volume in minute t , with the aggressor side taken directly from the Binance trade feed (avoiding the Lee-Ready (1991) algorithm). This measure is inspired by, but distinct from, the depth-based OFI of Cont, Kukanov, and Stoikov (2014), which is constructed from changes in best bid/ask depth in the limit order book.

Normalised OFI (NOFI). To facilitate comparison across periods of differing depth, we define $NOFI(t) = OFI(t) / (\text{BuyVolume}(t) + \text{SellVolume}(t))$, bounded in $[-1, 1]$, with values near plus (minus) one indicating minutes in which volume was overwhelmingly buyer- (seller-) initiated.

Forward returns. We construct three forward log-return series, $\text{ret_fwd_5m} = \log(\text{Close}(t+5)/\text{Close}(t))$ and analogously for 15 and 30 minutes, scaled to basis points in all reported regressions.

Funding rate. The Binance funding rate for XAUUSD is collected at its native four-hourly settlement frequency and merged onto the one-minute panel by a backward as-of join, so that each minute carries the most recently settled rate strictly available at that minute. The funding effects we report are therefore predictive, using only past-settled information, and are not a contemporaneous identity.

4.3 Session classification

We partition each trading day into four mutually exclusive and exhaustive intraday sessions by the UTC hour of each minute. Table 1 reports the definitions and observation counts, which sum to the full sample.

Session	UTC hours	Observations	Share
Asia	00:00-08:00	67,678	33.3%
Europe	08:00-14:00	50,760	25.0%
US	14:00-22:00	67,680	33.3%
Overnight	22:00-24:00	16,919	8.3%
Total	24 hours	203,037	100%

Table 1: Session classification and observation counts.

4.4 Descriptive statistics

Table 2 reports distributional properties of the two order-flow measures. OFI is near-symmetric and fat-tailed; NOFI is approximately mean-zero and bimodal, reflecting minutes dominated by one-sided flow. Appendix Figures A1 and A2 plot the full distributions and normal quantile-quantile diagnostics, confirming the heavy tails of OFI (excess kurtosis 6.4) and the bimodality of NOFI, whose mass concentrates near plus and minus one in fully one-sided minutes.

Variable	N	Mean	Std Dev	Skew	Kurt / Med.
OFI	203,037	-1.810	66.152	-0.305	6.406 / 0.000
NOFI	200,807	-0.000	0.513	0.024	-0.654 / -0.006

Table 2: Distributional properties of OFI and NOFI.

5. Methodology

Our baseline predictive regression for horizon h in $\{5, 15, 30\}$ minutes is

$$ret_fwd_h(t) = \alpha + \beta \cdot NOFI(t) + \gamma \cdot OFI(t) + \delta \cdot Volume(t) + \sum \theta(j) \cdot Session(j, t) + \phi_1 \cdot ret_1m(t) + \phi_2 \cdot ret_5m(t) + \varepsilon(t),$$

estimated by OLS with Newey-West (1987) heteroskedasticity- and autocorrelation-consistent standard errors (lags = 5); inference is unchanged under the Andrews (1991) automatic-bandwidth HAC estimator. Because the 5-, 15-, and 30-minute forward returns overlap, the regression error is serially correlated by construction (Hansen and Hodrick, 1980), which the HAC correction accommodates and which we further address through non-overlapping re-estimation in Section 7. To examine the funding channel we add the most recently settled funding rate and, crucially, its interaction with order flow:

$$ret_fwd_h(t) = \alpha + \beta_1 \cdot NOFI(t) + \beta_2 \cdot Funding(t) + \beta_3 \cdot [NOFI(t) \times Funding(t)] + controls + \varepsilon(t).$$

A non-zero β_3 implies that the informativeness of order flow is conditional on the funding regime. We also re-estimate the baseline within each session, split the sample by the sign and by quintiles of funding, and assess economic significance by sorting minutes into NOFI quintiles and comparing forward returns gross and net of realistic trading costs. Section 7 subjects the findings to an extensive robustness battery. All forward returns are expressed in basis points.

6. Results

6.1 The benchmark-anchored null: order flow is unconditionally uninformative

We begin with the unconditional benchmark, which the architecture of Section 2 predicts should be uninformative. Table 3 reports the baseline regression. The NOFI coefficient is economically negligible and statistically insignificant at every horizon ($t = 0.42, 0.19, -0.18$). The null is robust rather than fragile: it is unchanged across Newey-West lag lengths of 3, 5, and 10; the out-of-sample predictive R-squared is negative; rolling-window estimation finds significance in only about eleven percent of windows; and a leave-one-week-out jackknife leaves the full-sample coefficient near zero (mean $+0.016$, same sign in 71 percent of weeks). The only marginally non-zero baseline term is the contemporaneous OFI coefficient at the 5-minute horizon (-0.0015 , $t = -1.80$), which points to a very weak and non-robust short-horizon reversal. Minute-level trade imbalance is not, on its own, a stable predictor of short-horizon returns in this market (Figure 1). We treat this structured null, rather than any session anomaly, as the paper's first result.

Variable	β (5m)	β (15m)	β (30m)
const	0.138 (0.82)	0.858** (2.57)	1.747*** (3.42)
NOFI	0.023 (0.42)	0.016 (0.19)	-0.021 (-0.18)
OFI	-0.0015* (-1.80)	-0.0006 (-0.52)	0.0002 (0.09)
volume	0.0006 (1.19)	-0.0004 (-0.50)	0.0005 (0.47)
sess_asia	-0.162 (-0.90)	-0.665* (-1.82)	-1.563*** (-2.81)
sess_europe	-0.267 (-1.43)	-0.882** (-2.33)	-2.016*** (-3.54)
sess_us	-0.275 (-1.51)	-0.944** (-2.56)	-2.111*** (-3.77)
ret_5m	-255.2** (-2.08)	-278.5 (-1.37)	-295.9 (-1.12)
N	197,513	197,438	197,423
Adj. R ²	0.00134	0.00063	0.00069

Table 3: Full-sample baseline regression. Dependent variables in basis points; HAC-robust (Newey-West, lags = 5) t -statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

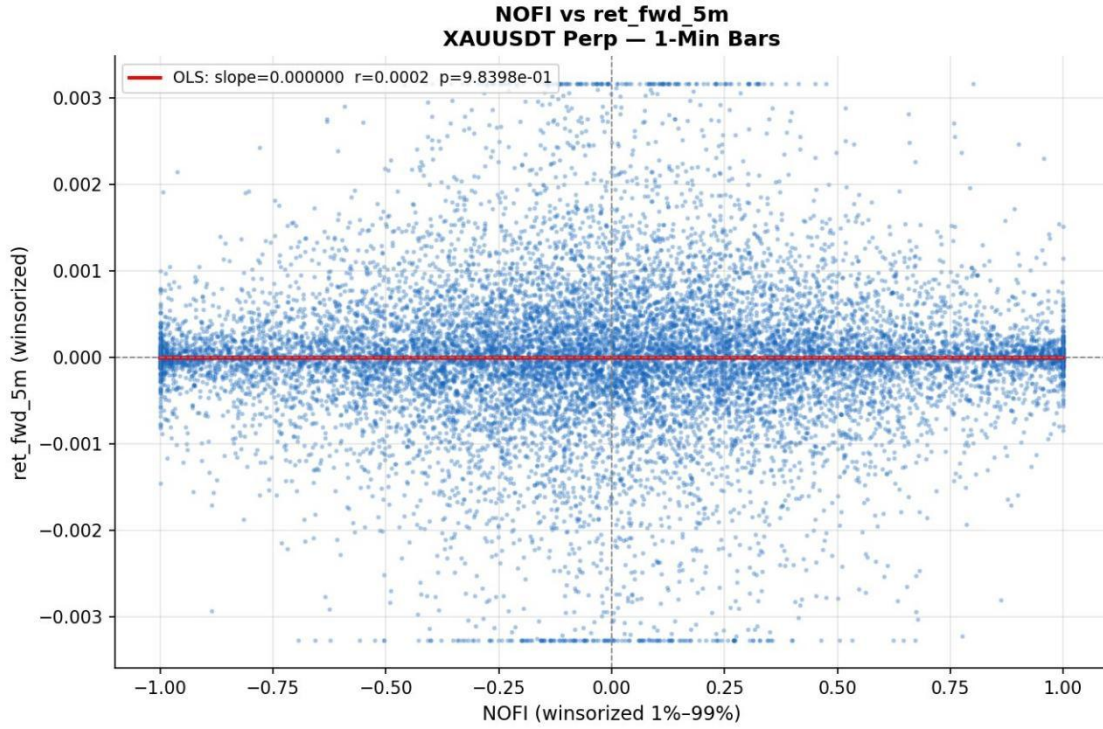


Figure 1: Raw bivariate NOFI vs 5-minute forward return; the full-sample structured null. Multivariate estimates with controls are in Table 3.

6.2 The funding channel: funding rates predict short-horizon returns

When the most recently settled funding rate is added to the baseline, it is positive and strongly significant at all three horizons (Table 4, row Funding: $t = 2.22, 2.95, 3.71$), while the NOFI main effect remains insignificant. The funding rate enters as a small decimal four-hourly rate (sample mean 0.7 basis points, standard deviation 5.0 basis points, capped at plus or minus two percent but rarely approaching that bound), so the raw per-unit coefficient is not directly interpretable. Standardised by the in-sample funding standard deviation, a one-standard-deviation increase in the funding rate raises the forward return by approximately 0.14, 0.38, and 0.71 basis points at the 5-, 15-, and 30-minute horizons — modest in absolute terms but estimated with high precision. As Section 7 shows, the funding effect is the single most robust relationship in the paper: it stays positive and mostly significant across leave-one-month-out, leave-one-week-out, extreme-event exclusions, and every alternative order-flow normalisation.

Term	5m	15m	30m
NOFI (β_1)	0.039 (0.71)	0.043 (0.52)	0.023 (0.19)
Funding (β_2)	278.7** (2.22)	754.2*** (2.95)	1404.7*** (3.71)
NOFI $\times$ Funding (β_3)	-263.2*** (-2.64)	-410.3** (-2.28)	-625.7** (-2.26)
N	197,513	197,438	197,423
Adj. R^2	0.00152	0.00105	0.00144

Table 4: Full-sample regression with the funding rate and its interaction with order flow. Baseline controls included but suppressed. Dependent variables in basis points; HAC-robust t -statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6.3 The conditionality of order flow on the funding regime

The same specification reveals the paper’s central mechanism. The interaction between order flow and funding is negative and significant at every horizon (Table 4, row NOFI x Funding: -263.2, -410.3, -625.7; $t = -2.64, -2.28, -2.26$). The informativeness of order flow is thus conditional on the funding state rather than constant. Before interpreting the interaction, we stress that it does not contradict the positive funding main effect of Section 6.2: the two coefficients capture distinct forces. The positive main effect reflects the persistence and directional component of the funding state itself — a high funding rate tends to accompany subsequent positive returns. The negative interaction is a marginal effect of a different kind — it describes how an additional unit of buying pressure is rewarded, conditional on that state. In standardised terms, a one-standard-deviation rise in funding shifts the NOFI slope by about -0.13, -0.21, and -0.31 basis points at the three horizons, enough to offset and reverse the small positive slope that prevails when funding is low. The sign has a direct positioning interpretation: when funding is positive — longs are crowded and pay shorts — a minute of strong buying pressure is more likely to represent over-extended positioning and is followed by lower returns, a reversal; when funding is negative, the same buying pressure is followed by continuation. Splitting the sample by the sign of funding confirms the pattern: the NOFI coefficient is negative throughout the positive-funding subsample and positive throughout the negative-funding subsample (Table 5). Although the subsample coefficients are individually imprecise, their signs are uniform across horizons and consistent with the interaction estimate, and a funding-quintile analysis shows the same gradient (positive NOFI slopes in the lowest-funding bins, turning negative in the highest). Because the funding rate enters as the most recently settled value (a backward as-of merge), the interaction is predictive rather than contemporaneous by construction, and lagging funding by one or two further settlement intervals preserves the negative sign at every horizon (Appendix Table A2). Figure 2 visualises both facets.

Funding regime	NOFI β (5m)	NOFI β (15m)/(30m)	N
Funding > 0 (longs crowded)	-0.082 (-0.94)	-0.176 (-1.44) / -0.276 (-1.60)	83,496
Funding < 0 (shorts crowded)	+0.178 (1.52)	+0.265 (1.29) / +0.153 (0.53)	26,423

Table 5: NOFI predictability by funding sign. HAC-robust t -statistics in parentheses.

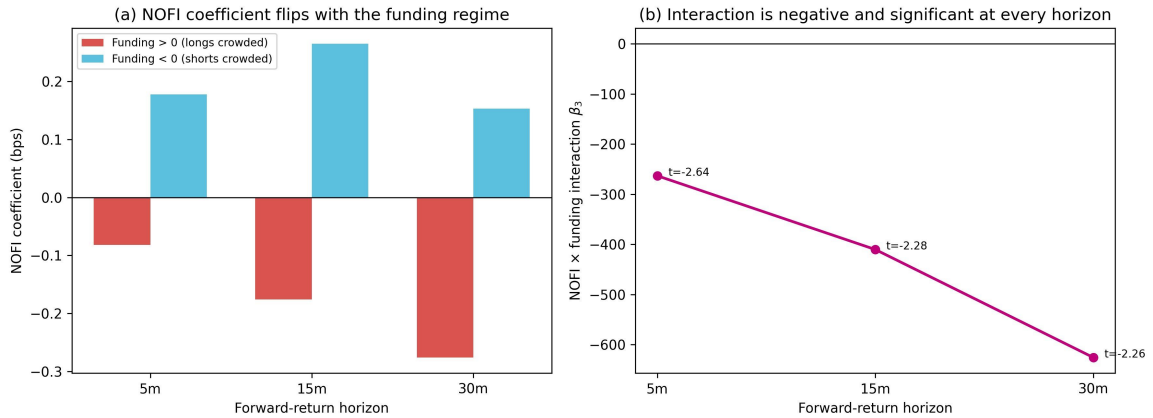


Figure 2: Order-flow predictability and the funding regime.

6.4 Session patterns: directionally stable but suggestive

Stratifying the baseline by session (Table 6) produces a positive NOFI coefficient in the European window at 15 minutes ($\beta = 0.290$, $t = 2.18$) and a negative coefficient in the overnight window at 30 minutes ($\beta = -0.653$, $t = -1.69$). We interpret these as suggestive rather than established. After correcting for the fifteen session-by-horizon tests, no session coefficient remains statistically significant: the p-value for the European 15-minute coefficient rises from 0.029 to 0.435 under the Benjamini-Hochberg adjustment, with Holm and Bonferroni corrections leading to the same conclusion. As Section 7 shows, the European coefficient weakens under shifted session boundaries, disappears under alternative order-flow normalisations, and changes sign under non-overlapping sampling; a fix-window placebo also fails to identify the London fix as the source of the pattern. The overnight coefficient is directionally stable but does not reach conventional significance and, given the fixed off-hours index and smoothed mark price documented in Section 2.3, may be partly mechanical. We therefore report these as directionally stable session-level patterns rather than definitive anomalies.

Session	β (5m)	β (15m)	β (30m)	p(15m) raw	p(15m) BH
Full sample	0.023 (0.68)	0.016 (0.85)	-0.021 (0.86)	0.850	0.903
Asia	-0.011 (0.90)	-0.023 (0.89)	0.124 (0.57)	0.886	0.903
Europe	0.080 (0.51)	0.290 (0.03)	0.151 (0.45)	0.029	0.435
US	0.085 (0.34)	-0.047 (0.77)	-0.077 (0.73)	0.772	0.903
Overnight	-0.149 (0.28)	-0.338 (0.18)	-0.653 (0.09)	0.185	0.903

Table 6: Session-stratified NOFI coefficients with multiple-testing correction. Parentheses report p-values, not t-statistics; the final columns report the raw and Benjamini-Hochberg-adjusted p-values for the 15-minute horizon. No cell is significant after correction.

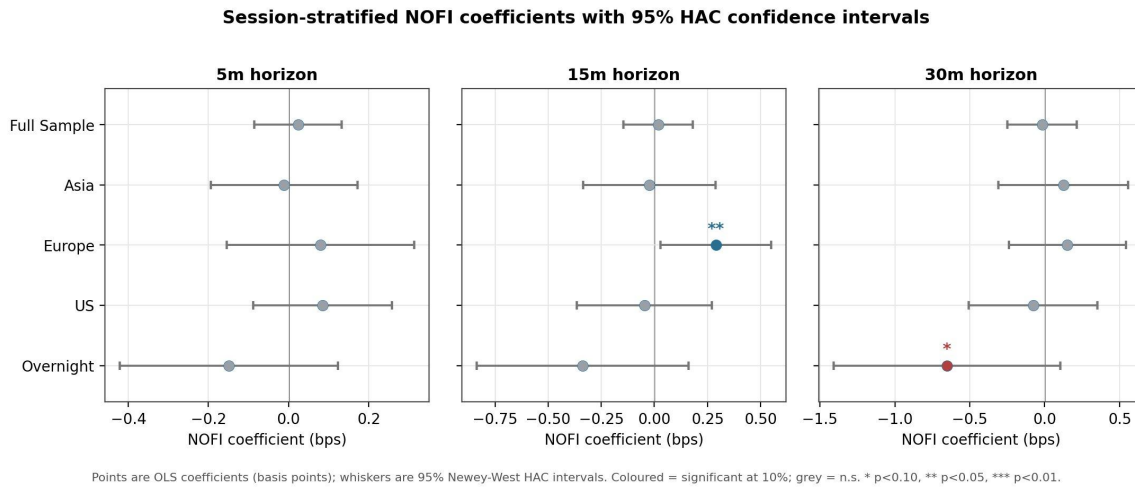


Figure 3: Session-stratified NOFI coefficients; see Table 6 for multiple-testing-adjusted inference.

6.5 No tradable alpha: the contribution is to market structure

Finally, we assess economic significance by sorting minutes into NOFI quintiles. The spread between the highest- and lowest-NOFI quintiles is small and, consistent with the reversal pattern above, negative: -0.198 bps ($t = -4.44$), -0.137 bps ($t = -1.75$), and -0.146 bps ($t = -1.29$) at 5, 15, and 30 minutes (Table 7). Against a conservative round-trip cost of about 14 bps (a 4.5 bps taker fee, a 1.0 bps half-spread, and 1.5 bps of slippage, applied at entry and exit), the net spread is roughly -14 bps at every horizon. Combined with the negative out-of-sample R-squared and the intermittent rolling-window significance documented above, this

establishes that order flow does not provide a directly tradable signal in this contract. We therefore interpret the paper’s findings as informative about the information structure of the market rather than about implementable trading alpha.

Horizon	Gross Q5-Q1 (bps)	t	Round-trip cost	Net (bps)	Tradable?
5m	-0.198	-4.44	14.0	-14.20	No
15m	-0.137	-1.75	14.0	-14.14	No
30m	-0.146	-1.29	14.0	-14.15	No

Table 7: Economic significance. Gross spread is the mean forward return of the top NOFI quintile minus the bottom quintile; cost assumptions stated in the text.

7. Robustness

Because the sample spans the contract’s first months of trading, we treat it as early-market evidence and address the concern that the results are sample-period specific with a battery of checks that hold the data fixed (Appendix Table A1). The extended tests reinforce the structured-null interpretation. The full-sample NOFI coefficient remains economically small and statistically insignificant across alternative Newey-West lag lengths, rolling windows, alternative normalisations, and extreme-event exclusions. Leave-one-month-out and leave-one-week-out tests show strong sign stability for the key European 15-minute, overnight 30-minute, and funding-rate coefficients, so the results are not driven by a single calendar month or isolated week: across the 21 weekly jackknife samples the European and overnight effects are 100 percent sign-consistent, and the funding effect is positive in every leave-one-out sample.

The funding-rate mechanism is the most robust relationship. Funding coefficients stay positive across leave-one-period-out and extreme-event tests and across all alternative order-flow normalisations, and the NOFI-by-funding interaction is significantly negative at all three horizons. The session-specific NOFI evidence, by contrast, should be interpreted cautiously. The European 15-minute coefficient is significant under conventional single-test inference ($p = 0.029$) but does not survive a multiple-testing adjustment across the session-by-horizon grid (Benjamini-Hochberg, Holm, and Bonferroni-adjusted $p = 0.435$); it weakens under shifted session boundaries (the t-statistic falling to about 1.1), loses significance under alternative NOFI normalisations such as a rolling z-score or rolling-volume scaling, and changes sign under non-overlapping (every-15-minute) return sampling. A randomised-NOFI placebo places the European 15-minute coefficient at the 7.4 percent tail of its permutation distribution — suggestive, but not decisive.

We also probe the funding interaction for residual endogeneity. The baseline uses the most recently settled funding rate (a backward as-of merge), so the interaction is predictive rather than a same-period identity. Lagging funding by one and two further four-hour settlement intervals leaves the interaction negative at every horizon — for example -453 ($t = -1.63$) at 30 minutes under a one-period lag and -294 ($t = -1.67$) at 15 minutes under a two-period lag — although its significance attenuates as the funding signal grows staler. This sign stability under additional lagging argues against a mechanical contemporaneous explanation and is consistent with the interaction reflecting the current state of leveraged positioning, which is most informative when funding is measured recently. Under non-overlapping (every-h-

minute) sampling the interaction is negative and significant at the 5-minute horizon (-379, $t = -1.96$) and negative though insignificant at 30 minutes, indicating that the short-horizon result is not an artifact of overlapping returns.

The benchmark-timing evidence does not support a fix-specific channel. In the placebo exercise, randomly allocated pseudo-fix windows reproduce the pre-fix order-flow coefficient in roughly half of all draws, and the event study around the London fixes does not single out the afternoon fix as a reliable predictor of returns. The timing also complicates a fix interpretation: under the fixed 08:00-14:00 UTC European-session definition, British Summer Time shifts the afternoon fix to the edge of, or outside, the session window. We therefore interpret the European-session coefficient as a weak institutional-time pattern rather than as evidence of a London PM-fix effect. Taken together, the robustness checks are consistent with the paper's main interpretation: order-flow informativeness in the XAUUSDT contract is conditional on the funding state, while session and benchmark-timing patterns remain suggestive rather than definitive. Earlier diagnostics on HAC lag length, subsamples, out-of-sample fit, and rolling windows, summarised in Appendix Table A1, point in the same direction.

8. Discussion

The empirical evidence is most consistent with a positioning-and-reversal mechanism rather than a classical informed-trading mechanism: in a contract tethered to an external benchmark, exchange-level order flow conveys information mainly about the state of leveraged positioning, which the funding mechanism prices, rather than about the fundamental value of gold, which is set elsewhere.

8.1 Implications for the market design of commodity-linked perpetuals

Read as a whole, the results connect three concrete design parameters of the contract to the way information enters its price. First, the funding cadence and cap. Because funding is the channel through which positioning information is priced, the four-hourly settlement frequency (rather than the eight-hourly cadence of major crypto perpetuals) and the per-interval cap determine how quickly that information is impounded and how far the perpetual can drift from its index before the restoring force binds; a faster cadence prices positioning sooner, which our funding-conditional results suggest is the economically relevant margin. Second, the off-hours index and mark-price construction. Holding the index fixed and smoothing the mark price when the underlying gold market is closed mechanically severs exchange-level flow from fresh benchmark information during those windows, so that overnight order-flow/return dynamics reflect exchange-internal inventory and liquidity rather than value discovery — a design choice that directly shapes where on the clock order flow can and cannot be informative. Third, leverage. Because leverage is what makes the funding rate a meaningful summary of crowded positioning, the available leverage governs the strength of the positioning-and-reversal channel we document. In each case a design parameter maps onto an information-transmission property, which is the sense in which our findings speak to the market design of this fast-growing contract class rather than to a tradable signal.

The overnight pattern should be read with the contract's off-hours pricing rules in mind. Because the index is fixed and the mark price smoothed when the underlying gold market is closed, overnight order-flow/return dynamics partly reflect exchange-internal inventory and liquidity rather than the impounding of fresh benchmark information; a purely adverse-selection reading would over-interpret the data. Finally, we emphasise that none of the documented effects is exploitable net of trading costs; the contribution is to market structure, not to alpha.

These patterns carry cautious, and mostly negative, implications for trading. A high reading of order-flow imbalance is not a buy signal: the quintile evidence shows that high-NOFI minutes are, if anything, followed by slightly weaker returns, and the sign of any order-flow effect depends on the funding state rather than on the direction of flow itself. The funding rate is the more informative state variable — crowded-long conditions warn of reversal in the wake of buying pressure rather than continuation — but the associated magnitudes are far too small to overcome trading costs. We therefore read the European-session tendency as, at most, a weak confirmatory feature within a broader positioning context, and the overnight tendency as a caution against chasing extreme order flow in thin, off-hours conditions, rather than as actionable signals.

9. Limitations and Conclusion

We provide the first high-frequency microstructure account of a commodity-linked cryptocurrency perpetual. Minute-level order flow does not predict returns on its own; its informativeness is conditional on the funding regime, and the funding rate is the active channel, while session effects are directionally suggestive but not robust and no effect is tradable net of costs. The study has clear limitations. The sample covers a single venue, a single contract, and the instrument's first months, so the estimates are early-market evidence; we mitigate, but cannot eliminate, sample-period concerns through the robustness battery. We do not yet control for external gold-market variables — spot gold, COMEX gold futures, the dollar index, and scheduled macro announcements — and merging these series is a natural next step that would sharpen the separation of benchmark-driven from positioning-driven effects. The cross-asset extension to the companion silver perpetual (XAGUSDT) is also immediate. These extensions would strengthen the external validity of the results, but the central finding is already clear: in a benchmark-anchored commodity perpetual, the funding mechanism is the state variable through which leveraged positioning becomes informative for short-horizon returns, while raw minute-level order flow is not, on its own, a reliable signal.

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Data Availability Statement

The trade and funding data underlying this study were collected from the public Binance Futures API; London gold fix reference times were obtained from the public LBMA feed. The derived one-minute panel, the variable-construction pipeline, and the estimation and robustness code are available from the author on reasonable request. No proprietary or licence-restricted data were used.

Appendix A. Sample-period and specification robustness

Table A1 summarises the robustness battery referenced in Section 7. Each row re-estimates the four core effects under a perturbation that holds the data fixed. Entries are coefficients with HAC-robust t-statistics in parentheses.

Perturbation	Full NOFI 15m	Europe NOFI 15m	Overnight NOFI 30m	Funding 15m
Baseline	+0.016 (0.19)	+0.290 (2.18)	-0.653 (-1.69)	+754.0 (2.95)
Leave-out Jan	+0.121 (1.34)	+0.244 (1.45)	-0.414 (-0.98)	+1052.7 (3.15)
Leave-out May	-0.011 (-0.12)	+0.330 (2.24)	-0.720 (-1.64)	+755.3 (2.93)
Jackknife mean (21 wks)	+0.016	+0.291	-0.653	+758.9
Jackknife % same sign	71.4%	100%	100%	100%
Europe window 07-13	+0.014 (0.17)	+0.173 (1.15)	-0.331 (-1.13)	+753.3 (2.94)
Europe window 09-15	+0.016 (0.19)	+0.146 (1.09)	-0.653 (-1.69)	+753.7 (2.95)
Drop top-1% ret_1m	-0.039 (-0.56)	+0.164 (1.30)	-0.639 (-1.78)	+630.1 (2.71)
Drop top-5 vol. days	+0.069 (1.04)	+0.282 (2.40)	-0.566 (-1.49)	+396.5 (1.83)

NOFI z-score	-0.031 (-0.37)	-0.007 (-0.05)	-0.020 (-0.06)	+753.7 (2.95)
NOFI rolling-scaled	-0.016 (-0.33)	+0.051 (0.72)	-0.270 (-1.17)	+753.8 (2.95)
Non-overlapping returns	+0.149 (0.42)	-0.340 (-0.65)	-1.712 (-1.34)	-

Table A1: Robustness battery. The full-sample null and the funding effect are stable across all perturbations; the European and overnight effects are directionally stable but lose significance under alternative normalisations and non-overlapping sampling. Multiple-testing correction is reported in Table 6; placebo tests are described in Section 7.

Fix-window event study (identification check). An event study around the London fixes does not support a fix-driven channel. Around the afternoon (PM) fix, the 15-minute NOFI coefficient is insignificant in three of four windows and only marginally significant in the $[-30,0]$ window, where it is negative ($\beta = -0.908$, $t = -1.77$) — the opposite sign to the positive European-session coefficient — while all four morning-fix windows are insignificant. Together with the fix-window placebo, in which randomly placed pseudo-fix windows reproduce the pre-fix coefficient in roughly half of draws, this indicates that the European pattern is not attributable to the LBMA fix.

Appendix A2. Endogeneity and overlapping-return checks for the funding interaction

Table A2 reports the NOFI-by-funding interaction coefficient under additional funding lags and non-overlapping sampling. The negative sign is preserved at all horizons under both one- and two-period lags; significance is strongest in the baseline (as-of) specification and attenuates as funding becomes staler. Non-overlapping sampling preserves a significant negative interaction at the 5-minute horizon.

Specification	β_3 (5m)	β_3 (15m)	β_3 (30m)
Baseline (most recently settled funding)	-263.2*** (-2.64)	-410.3** (-2.28)	-625.7** (-2.26)
Funding lagged +1 settlement (4h)	-165.2 (-1.48)	-288.1 (-1.45)	-453.5 (-1.63)
Funding lagged +2 settlements (8h)	-134.1 (-1.35)	-294.0* (-1.67)	-252.1 (-0.97)
Non-overlapping sampling	-379.2** (-1.96)	+149.8 (0.26)	-1410.5 (-1.37)

Table A2: NOFI-by-funding interaction under additional funding lags and non-overlapping sampling. Dependent variables in basis points; HAC-robust t-statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Appendix A3. Variable distributions and correlations

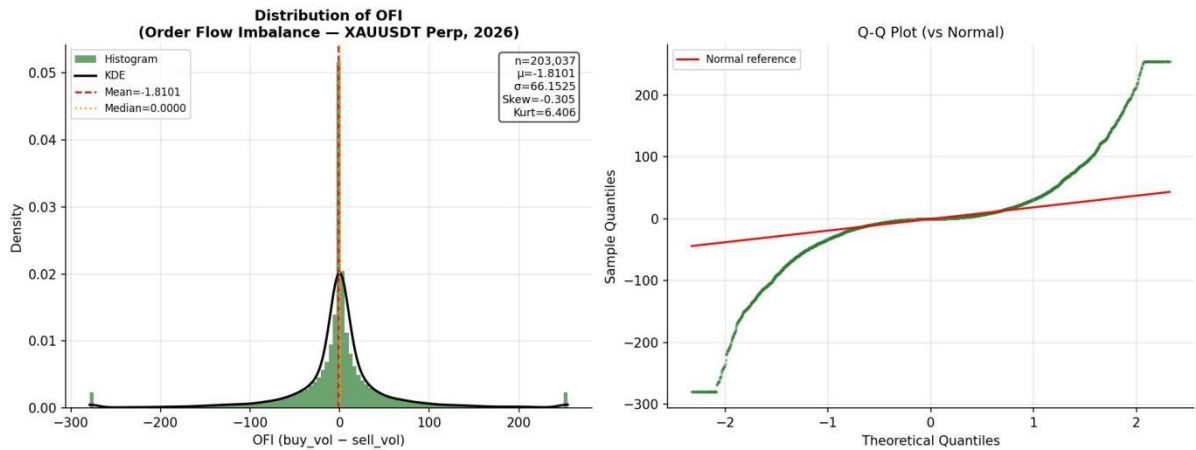


Figure A1: Distribution and normal Q-Q plot of OFI ($n = 203,037$). OFI is near-symmetric (skew -0.31) with heavy tails (excess kurtosis 6.4).

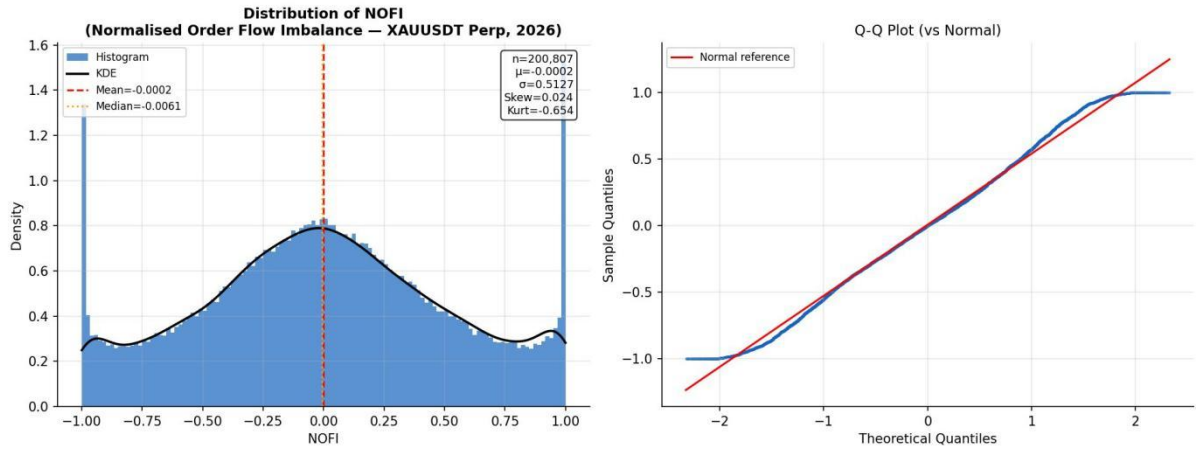


Figure A2: Distribution and normal Q-Q plot of NOFI ($n = 200,807$; mean -0.0002 , sd 0.513). NOFI is bimodal, with mass at plus and minus one corresponding to fully one-sided minutes.

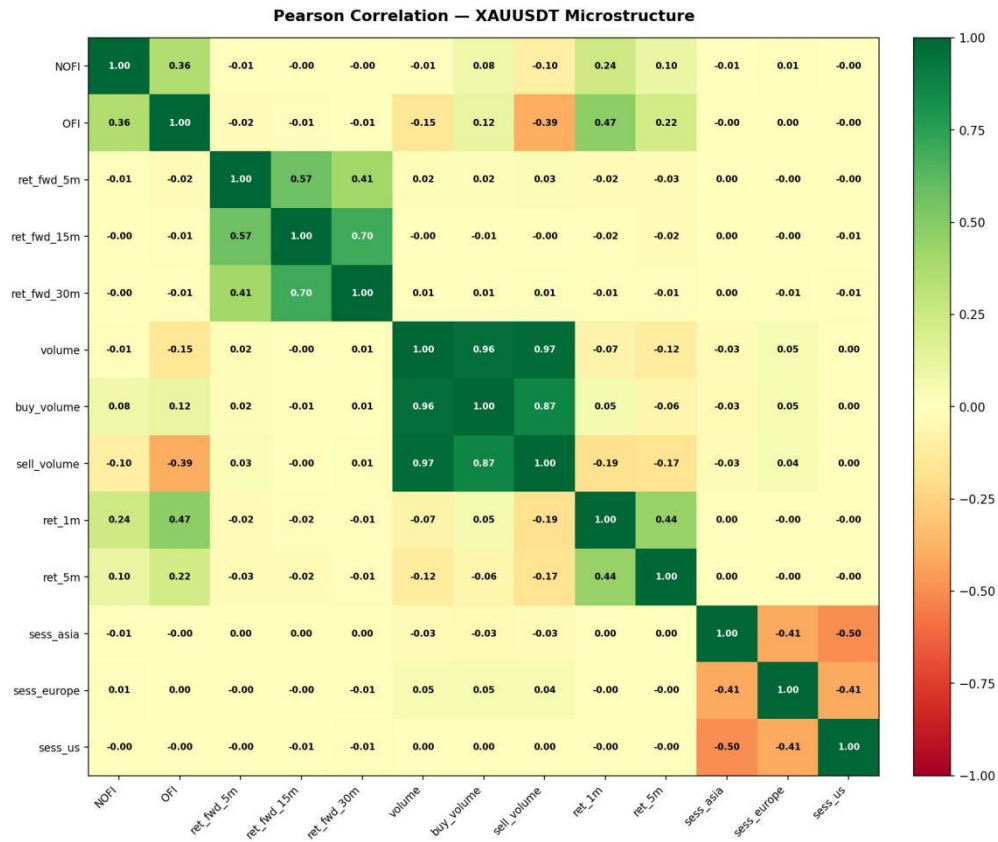


Figure A3: Pearson correlation matrix of the main variables. NOFI and OFI are essentially uncorrelated with all forward returns (absolute r at most 0.02), consistent with the full-sample null, while volume, buy-volume, and sell-volume are highly collinear, motivating the use of net imbalance rather than separate buy and sell terms. The Spearman matrix is qualitatively identical and available on request.

The bimodality of NOFI documented in Figure A2 is a feature of low-volume minutes rather than a construction artifact. Figure A4 splits the sample into volume terciles: the mass at the boundaries is concentrated in the low-volume tercile (34.0% of those minutes have absolute NOFI above 0.8), falls to 10.7% at the middle tercile, and to 1.9% in the high-volume tercile, where the distribution is approximately bell-shaped. Because the main analysis uses the net imbalance measure and yields a structured null, and the boundary mass is confined to low-volume minutes that carry little economic weight, the bimodality is a descriptive feature of the data rather than a driver of the results.

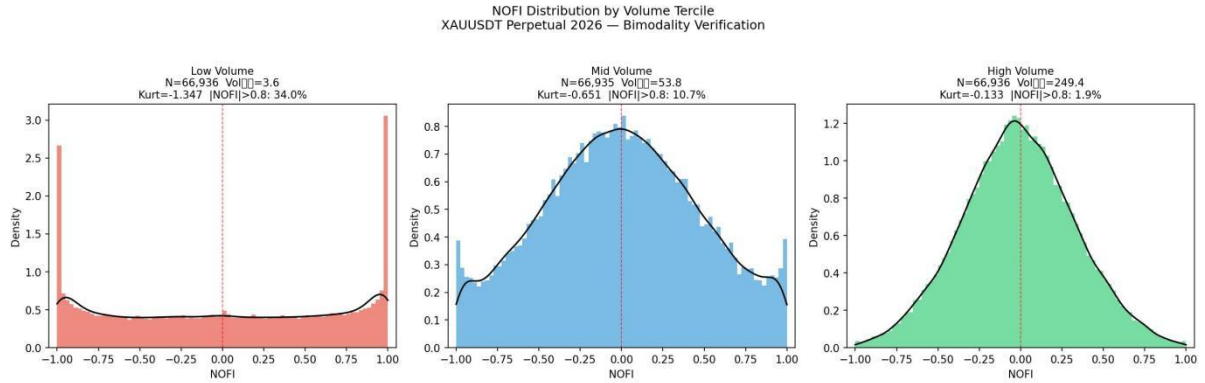


Figure A4: NOFI distribution by volume tercile. The boundary mass at plus and minus one is concentrated in low-volume minutes ($|NOFI| > 0.8$ in 34.0% of low-, 10.7% of mid-, and 1.9% of high-volume minutes), confirming that the bimodality is a low-volume feature rather than a construction artifact.

Finally, Figure A5 plots the 30-day rolling NOFI coefficient with 95% HAC confidence intervals at the 5- and 15-minute horizons. The coefficient hovers around zero throughout the sample and is individually significant in only a small minority of windows (about eleven percent), with the rolling R-squared remaining near zero. This temporal view reinforces the full-sample structured null: there is no stable window in which raw order flow predicts returns.

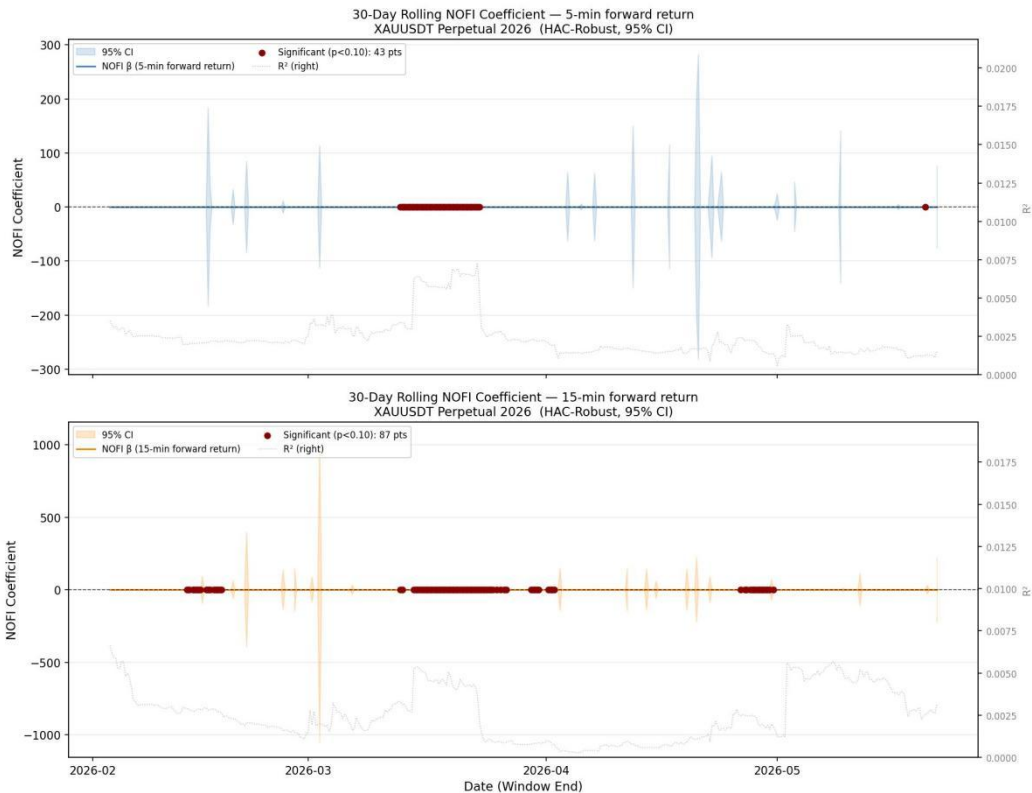


Figure A5: 30-day rolling NOFI coefficient (HAC-robust, 95% CI) at the 5- and 15-minute horizons, with rolling R-squared on the right axis. Significance appears in roughly eleven percent of windows; the coefficient is otherwise indistinguishable from zero.